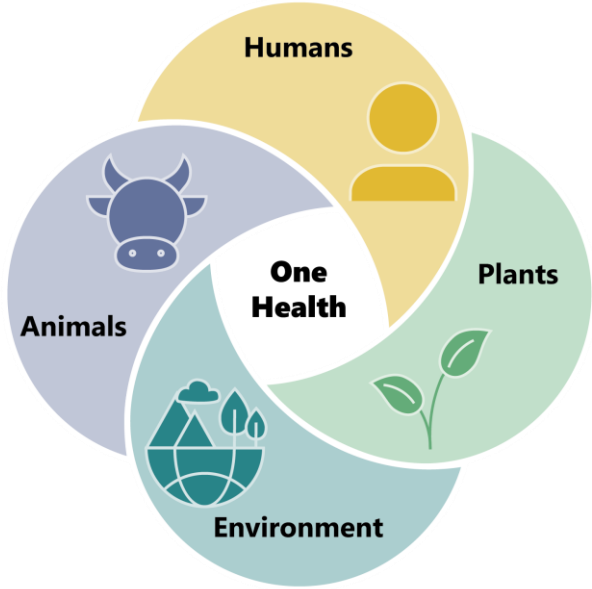
Simulation studies in regulatory science
ISBS 2026



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The contents of this presentation are my personal opinion. My remarks do not necessarily reflect the official view of AGES, BASG, EMA, or any associated working party or committee.

Settings for Simulation Studies

From Method Development to Application

- Method evaluation / neutral comparison
  - Compare available methods and designs
  - Not specific to one trial or development program
  - Comparison of multiple operating characteristics across many scenarios
- Evaluation of methods proposed for regulatory use
  - Evaluate methods in scenarios not considered in initial publication
  - Compare performance of methods published separately
- Demonstrating operating characteristics
  - Demonstrate sufficient performance in relevant operating characteristics
  - Specific to one trial

Research Project:

Use of causal inference methods commonly used in non-interventional studies to estimate treatment effects in clinical trials

Causal Inference in RCTs project overview

Project & Consortium

- EMA Tender EMA/2020/46/TDA ROC 36, Lot 3: Statistical research
  "***Use of causal inference methods commonly used in non-interventional studies to estimate treatment effects in clinical trials.***"
- **CONFIRMS: CONsortium For Innovation in Regulatory Medical Statistics**
  - Center for Medical Data Science (CeDAS),
    Medical University of Vienna (MUW), Austria
  - Department of Medical Statistics,
    University Medical Center Göttingen (UMG), Germany
  - Pharmacometrics Research Group,
    Uppsala University (UU), Sweden
  - Austrian Medicines and Medical Devices Agency (MEA),
    A Business Division of the Austrian Agency for Health and Food Safety (AGES), Austria

Methdological Approach

Simulation Study Informed by Literature Review

- Screen the scientific literature and regulatory documents for:
  - Randomized clinical trials in which causal inference methods have been
  - Methodological approaches to address intercurrent events based on causal inference
- In the screened documents identify:
  - Scenarios in which the causal pathway from randomized treatment to outcome may be affected by intercurrent events
  - Causal inference methods suitable for testing and estimating treatment effects that appropriately account for these intercurrent events
- Assess the statistical properties of the selected methods in a simulation study based in varying scenarios
- Simulation Study using Clinical Scenario Evaluation[1] and ADEMP[2] Frameworks

Simulation Study Protocol (1)

Selection of Scenarios and Methods

- Relevant and representative scenarios
  - Cover most important estimand strategies
  - Allow for application of a range of causal inference methods
- Parameters of data generation
  - Realistic, based on review of literature and regulatory documents
  - Covering a broad range
  - One core scenario + vary parameters one at a time
- Methods
  - Methods proposed in literature and used in regulatory proceedings
  - Reference methods used in similar scenarios without explicitly addressing causal inference

Simulation Study Protocol (2)

Scenarios, Estimands and Methods

Type of ICE and endpoint	Summary measure	Strategy	Methods
Oncology setting with switching and time to event endpoint	Hazard ratios	Hypothetical	RPSFTM IPW TSE g-computation
Diabetes trial with rescue medication and continuous endpoint	Difference in means	Treatment policy	IPW
		Hypothetical	IPW de-mediation g-computation
Vaccine trial with incomplete vaccination (lack of tolerability) and binary endpoint	Vaccine efficacy	Principal stratum of compliers with vaccination regimen	IV Propensity score
Safety trial with discontinuation of treatment and time to event endpoint	Hazard ratios	While on treatment	Competing risk IPCW

Simulation Study Protocol (3)

Simulations and Performance Metrics

- 10000 Replications (may be reduced if not feasible)
  - Numbers of bootstrap samples and multiple imputations could be reduced if higher number not feasible
- Performance Metrics
  - power (at one-sided alpha=2.5%)
  - Type 1 error rate (at one-sided alpha=2.5%)
  - Bias, variance, MSE
  - Coverage and average width of 95% CI
- Analysis
  - For power and Type 1 error rate, coverage probabilities: exact binomial CI
  - For bias: normal assymptotic CI

Programming & Running of Simulations

Software Development Approach

- Package structure setup
  - R package, unit tests, version control, dependency management
  - Functions shared across scenarios
  - Template, vignettes and documentation to ensure consistency and compatibility with SimDesing package
- Programming tasks were divided among teams for scenarios
  - Use of feature branches
  - Automated unit tests, review & signoff before merging into main branch
- Simulations run separately for each scenario
  - Different requirements on computing capacity
  - Iterate quickly and run locally where possible, submit to cluster if heavy lifting is need
  - Run in batches and aggregate results

Presentation, Dissemination

Publication & Reproducibility

- Deliverables to EMA:
  - Study Protocol and Study Report published in the EMA real world data catalog
  - Webinar for European medicines regulatory network
- Paper:
  - In previous project: good interest in literature review as well as in simulation study in terms of acceptance by journals as well as citations
  - Upcoming paper on simulation study of current project
- Simulation Code:
  - R-package + simulation scripts on github
  - In previous project also published in CRAN, how relevant is functionality outside of project?

Learnings, Simulation Studies

Planning Simulation Studies

- Cover relevant scenarios!
- Parametrisation
  - Interpretable: regression parameters, summary measures or effect sizes?
  - Vary parameters individually, avoid complex dependencies between parameters
    - Example: effect size and sample size
- For simulation & for presentation:
  - pick relevant parameter values or all combinations from full factorial design?
- Present selected representative scenarios or range of parameter values?
  - Pick representative scenario or worst case across parameter values
  - When presenting all: nested-loop plots[3], interactive presentation (e.g. Shiny App)

Learnings, Implementation

Programming and Conducting Simulation Studies

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- Test code with medium number of replications, some errors might only show up with low probability or for some combinations of parameters
- Optimize “low hanging fruit” where large speedups are possible with little work
- Time simulations during testing / development and plan accordingly for the final simulations
- What can be left out if it’s not feasible? Pre-specify, discuss with sponsor:
  - Only assess some operating characteristics (bias and variance only, leave out CIs where bootstrapping would be necessary)
  - Reduce bootstrap samples, imputations, …
  - Leave out irrelevant parameter values, run simulations on a sparser grid, reduce number of replications

Examples:

Evaluation of methods proposed for regulatory use

Covariate Adjusted Log-Rank Test

Evaluation of a Proposed Test

- 2024 Paper by Ye et.al. proposing the test[4] included asymptotic properties and some simulations
- Interest by regulators due to new properties: marginal estimate + stratification, potential gains in precision
- Uncertainties about small sample properties, performance in sequential designs, … remained
- Additional simulations filling the gaps and informing discussions on when the proposed test can be easily accepted and when scrutiny is necessary [5]

Similarity Criteria in Quality Comparisons

Comparison of Methods Used

- Multiple criteria proposed or used, based on statistical considerations, heuristics, …
- Many papers comparing and proposing methods [6,7,8]
  - Evaluation of performance with respect to different similarity conditions[9]
  - Role of variation differs depending on similarity condition it is a nuisance parameter only relevant for estimation or relevant for the condition
  - Different performance measures of similarity criteria (Average t1e, kappa-cover, …)
- Need for broader comparison & simulation studies
  - More methods in the same comparison
  - Multiple relevant similarity conditions
  - Relevant performance metrics → relevant for parametrisation and parameter space
- High relevance due to tailored approach[10]

Regulatory Use and Upcoming Guidance:

Demonstrating operating characteristics

Demonstrating Operating Characteristics

In Regulatory Interactions

- Routinely used for power & sample-size calculations
- For complex designs or methods, when analytical proof of operating characteristics is not available
- Few scenarios
  - Scenarios under the null-hypothesis to estimate Type 1 error rate
  - Violation of assumptions, e.g: non-exchangeability of borrowed and internal data
- No overall Guidance on Simulation Studies, but references in methodological draft guidelines
  - ICH E20 Guideline on adaptive designs for clinical trials (Step b., Draft Guideline adoption by Regulators) [11]
  - FDA: Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (Draft Guidance for Industry) [12]

Simulations in ICH E20

Simulations in the Draft Guideline on Adaptive Designs for Clinical trials

- Define Objectives of Simulation Study
- Adress key operating characteristics (Type 1 error rate, expected sample size, …)
  - Report average but variability, minimum, maximum might also be relevant
- Scenarios should cover the plausible range of assumptions
  - Treatment effects and nuisance parameters
  - Operational assumptions
  - Justified based on clinical and statistical considerations
- Describe and justify implementation details
  - Data generation, replications
- Document design, results and conclusions of the simulation study

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Protocol, study report,
R-package, paper:

https://ci-rct-sim.github.io/CI.RCT.links/

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CONFIRMS Projects

EMA Tender Projects by the CONFIRMS Consortium

- Statistical methods for time-to-event endpoints with non-proportional hazards in clinical trials pivotal for benefit risk decision making (2022-2024)
- Uncertainty quantification for complex models supporting regulatory decision making (2025-2026)
- Use of causal inference methods commonly used in non-interventional studies to estimate treatment effects in clinical trials (2025-2026)
- Characterisation of complex clinical trials in the European regulatory context (2025-2026)

Simulation Study Frameworks

ADEMP and CSE



ADEMP	CSE
Aims	
Data-generating mechanisms	Assumptions
Estimands	Options
Methods	
Performance-measures	Metrics

Used Tools

Tools for Simulation Studies in R

AGES

- Packages and Tools used for the project:
  - **Simulation Packages:** SimDesign, SimNPH (own package developed for a previous project)
  - **R package-development:** devtools, usethis, roxygen2
  - **Dependency management:** renv
  - **Unit-Tests:** testthat, covr, usethis, github-action r-lib/R-CMD-check
  - **Project management:** github projects, github issues, pull-requests
  - **Parallelisation**: parallel, mirai, slurm
- Useful resources:
  - CRAN Task View: Package Development and Maintenance
  - CRAN Task View: High-Performance and Parallel Computing with R